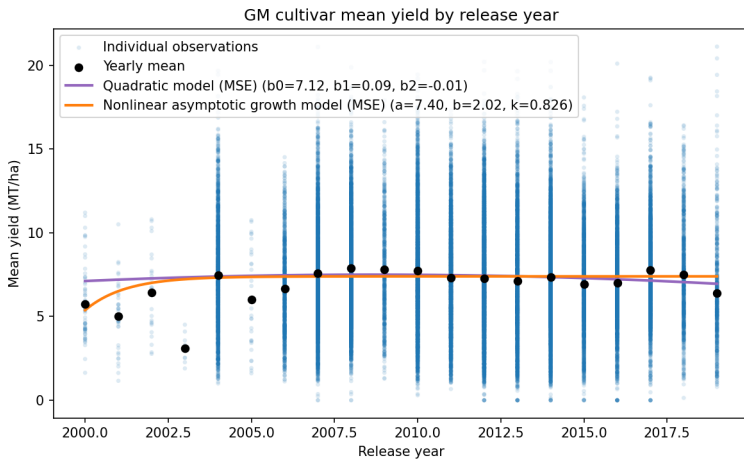


How much has the yield of GM maize changed over the years?

Plots of both models on the dataset



Confidence intervals

```
quad confidence interval (b0, b1, b2):  
[[ 6.86235047e+00  7.34378031e+00]  
 [ 4.52926157e-02  1.34242200e-01]  
 [-7.17067509e-03 -3.26615519e-03]]  
nonlinear confidence interval (a, b, k):  
[[7.36352838 7.42922032]  
 [1.47146106 2.5629805 ]  
 [0.68143726 1.01087891]]
```

```
gain quad confidence interval: [-0.35540478  0.04937591]  
gain nonlinear confidence interval: [1.47146098 2.56298047]  
gain difference confidence interval (quad - nonlinear): [-2.70658151 -1.6885479 ]
```

Interpretation

Both models show yield increasing in the early years, but the quadratic rise is smaller compared to the non linear model. Afterwards the non linear model remains flat (which is to be expected for an asymptotic growth model).

The models do not agree: the 95% bootstrap confidence interval for the 2000–2019 yield gain is $[-0.36, 0.05]$ for the quadratic model but $[1.47, 2.56]$ for the saturating model. The confidence interval for the difference is $[-2.71, -1.69]$. The models mainly disagree between 2000 and 2004, since there is less data in comparison to later years.